

AudioAware: AI-Guided Assistant Device for the Deaf and Hearing Impaired

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Abstract

AudioAware, an accessibility device for the deaf or hearing impaired, will accurately localize and identify sound sources in real time. This study will fill an important gap in current research by leveraging machine learning techniques to target the underprivileged community. The system includes three main components.

- **JerryNet**: A custom designed CNN architecture that determines the direction of arrival (DoA) for *nine possible directions*. The input is a phase matrix generated by the concurrent audio of four different microphones.
- **Audio Classification Model**: This model uses only audio data to fine-tune the Contrastive Language–Audio Pretraining (CLAP) to identify the exact sound classes.
- **Multimodal Integration Model**: An accurate sound localization model that combines audio, visual, and text data to locate the exact sound sources in the images. The model consists of two modules:
 - **Object detection** using Yolov9 to generate all the objects’ bounding boxes.
 - **Audio-visual localization model** (SIRA-SSL) identifies the optimal bounding box using complete Intersection over Union (CIoU).

The hardware consists of four-microphone rectangular formation and a camera mounted on a glasses with a wristband for displaying necessary information like direction.

Results: On a custom collected data set, JerryNet achieved a precision of 91. 1% for the sound direction, outperforming all the baseline models. The CLAP model achieved 98.5% and 95% accuracy on custom and AudioSet datasets, respectively. The audio-visual localization model within component 3 yielded a cloU of 0.892 and an AUC of 0.658, surpassing other similar models.

Introduction

Product	Total Cost
 Cochlear Implant	\$50000
 AudioWare	\$30

Utilizing multiple microphones and cameras, it employs a multimodal approach to assistive technology, enabling sound localization and audio classification. The system consists of smart glasses with audio-visual input and a wristband for user display, integrating these functions into a seamless control loop for reliability.

Beyond this device, the software components can be adapted for broader applications, contributing to a growing ecosystem of assistive technologies that enhance quality of life. Further details on functionality and system design will be explored in later sections.

Research Questions

How to identify the exact location and classify a sound source through a combination of audio and visual input?

Data Collection

A custom dataset consisting of exactly 450 sets collected from four concurrent microphone. There are 50 data for each of the nine distinct directions of arrival. The data is distributed between indoor and outdoor environments to increase model accuracy for different background noise. The data is intentionally designed so the model can be trained to isolate important sounds even in noisy conditions. The audio input from the four microphones is amplified to exaggerate the volume difference for a more significant result.

Hardware + Test Scenarios



Photo of Homemade Glasses. Produced by student.

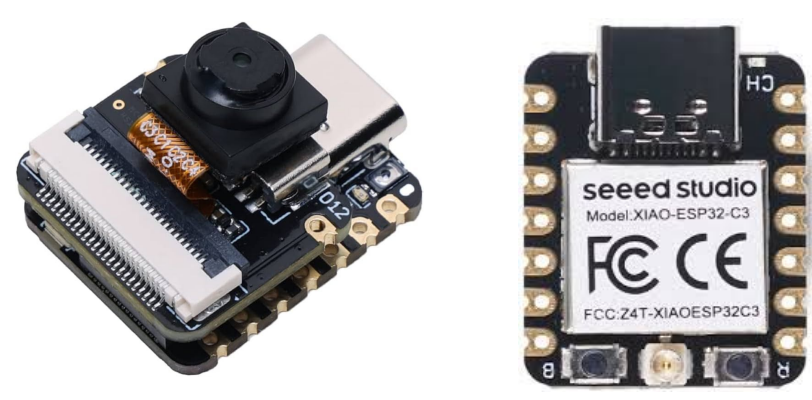


Image of Xiao esp32. Each equipped with a microphone and camera. Published data from amazon.com



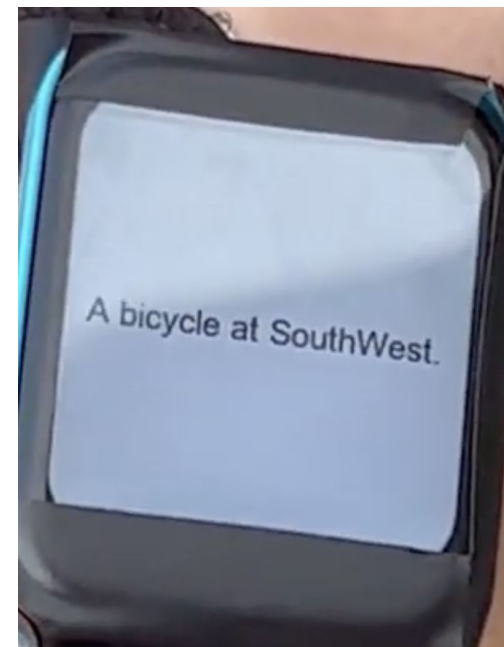
Photo of Homemade Glasses. Produced by student.



ESP32S3



Example test scenario with a bike bell. Produced by the student.



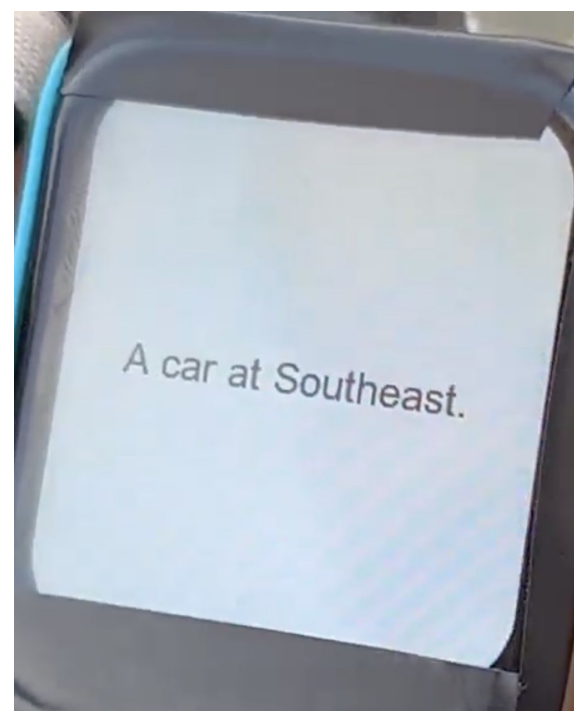
The text display for wristband showing the exact object making the sound and the direction of the sound. Produced by the student.



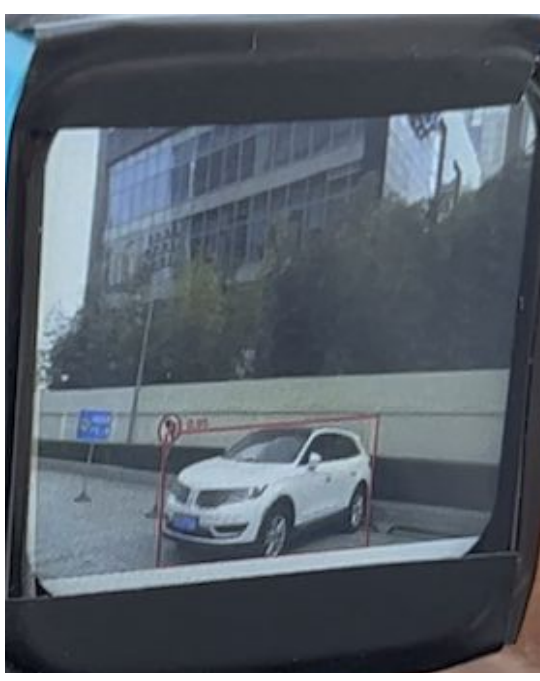
The image display for wristband showing a box of the exact object making the sound, which is the bicycle. Produced by the student.



Example test scenario with a car honking. Produced by the student.



The text display for wristband showing the exact object making the sound and the direction of the sound. Produced by the student.

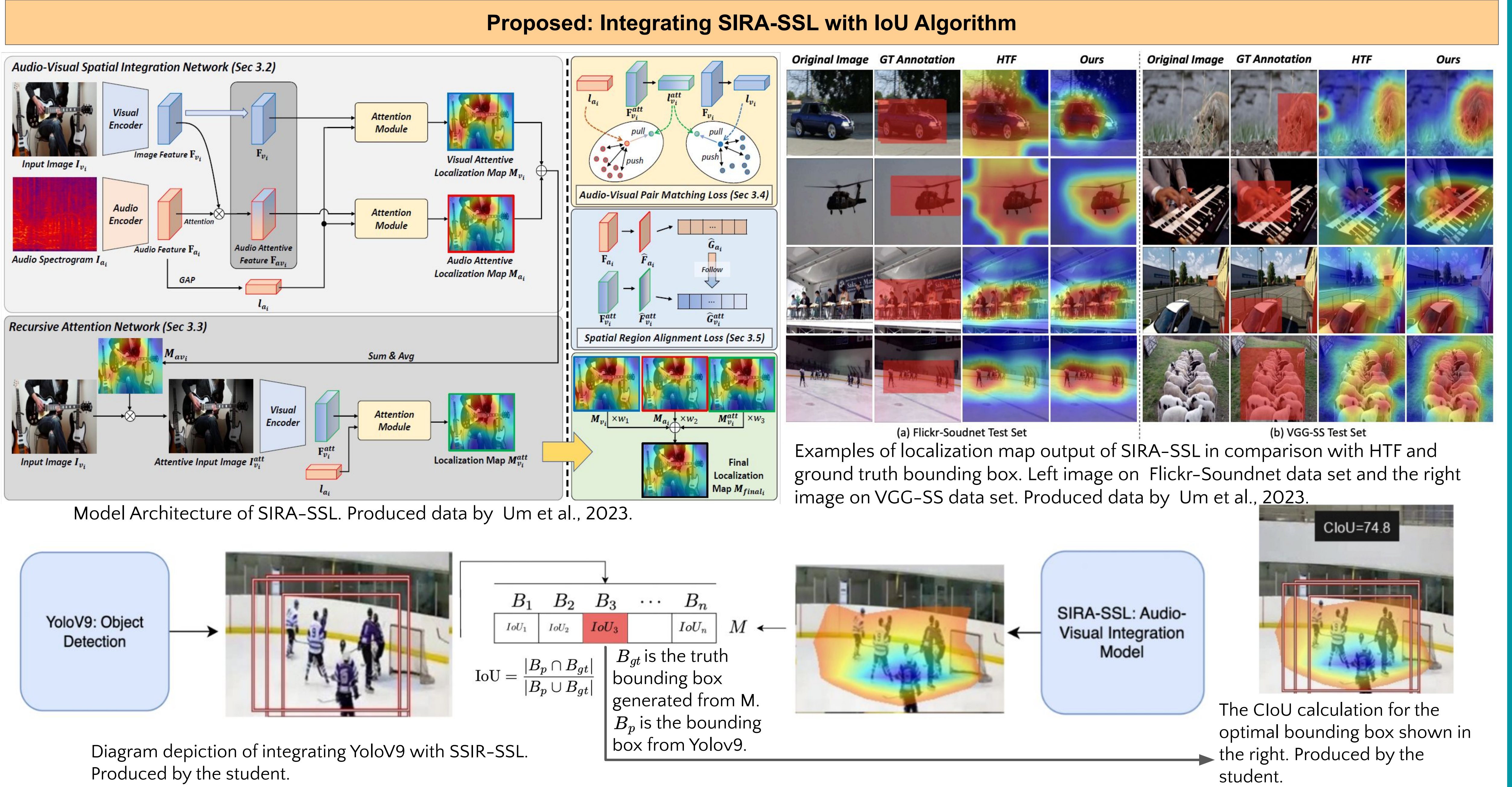
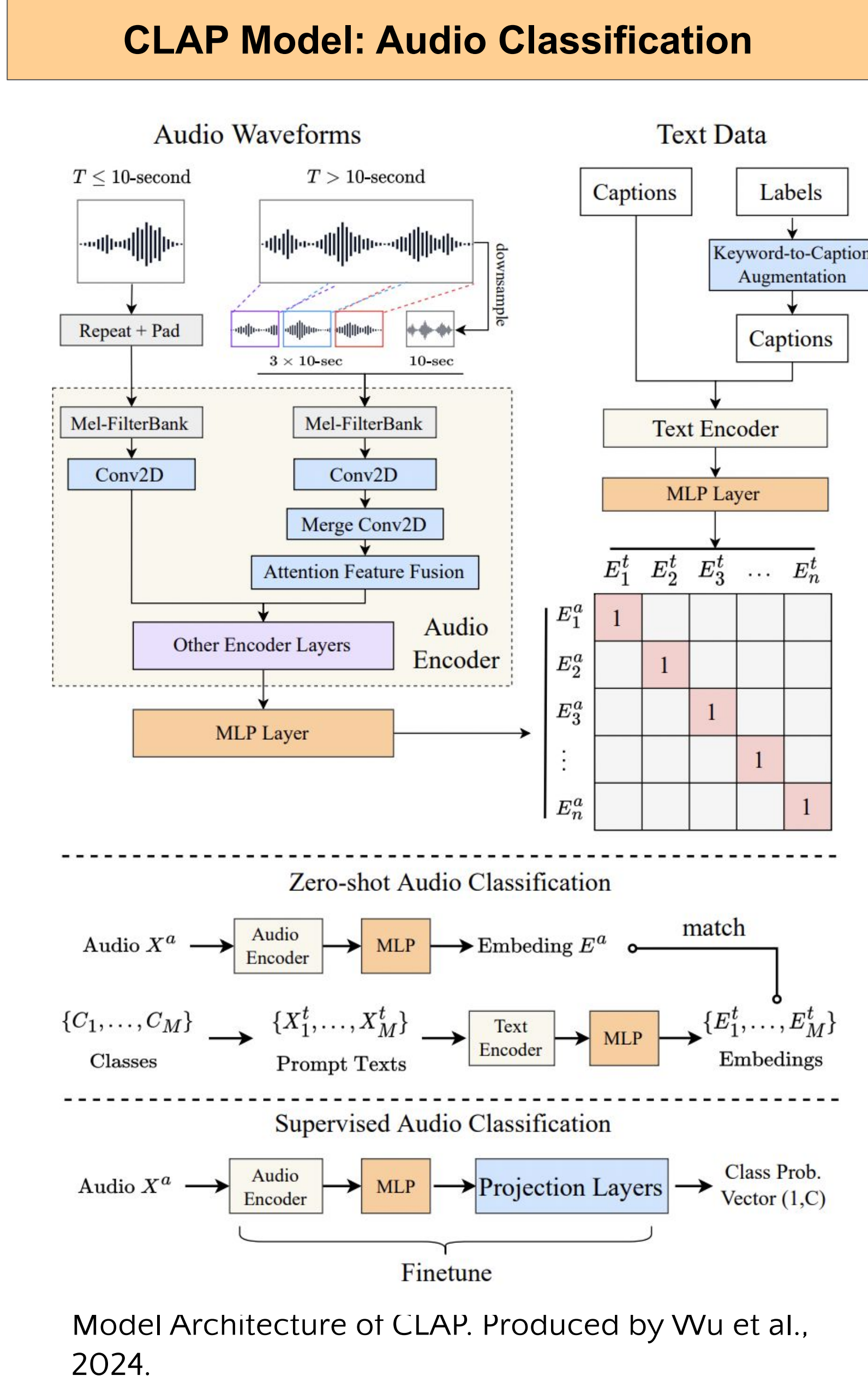
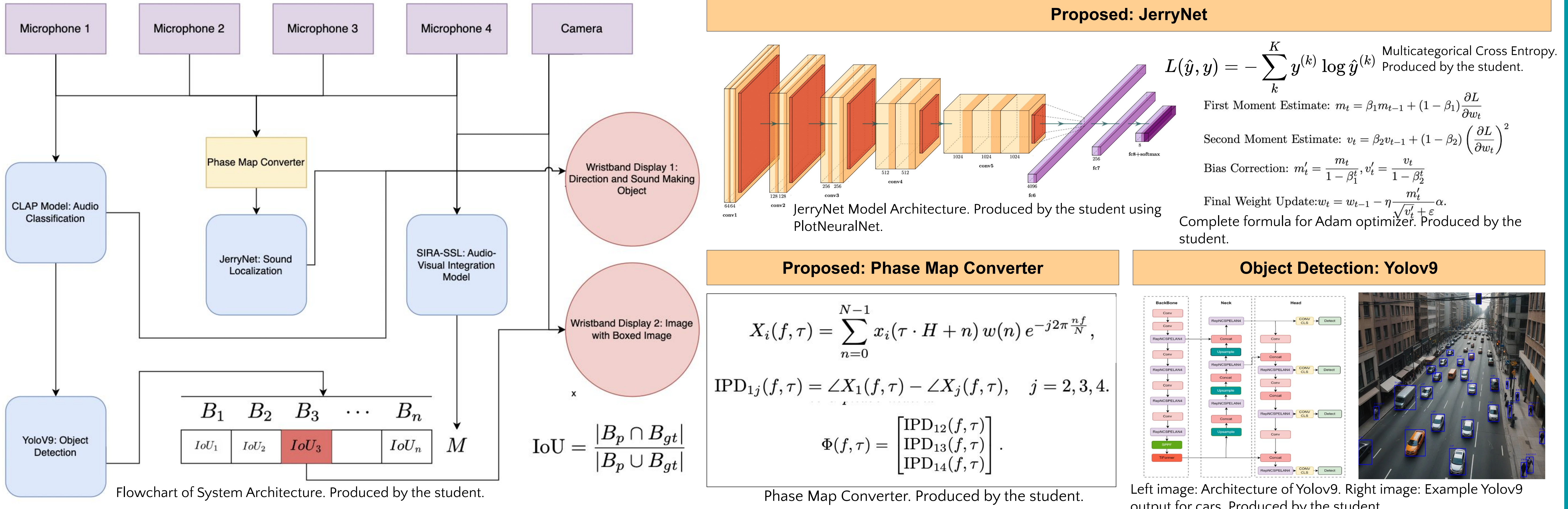


The image display for wristband showing a box of the exact object making the sound, which is the car here. Produced by the student.

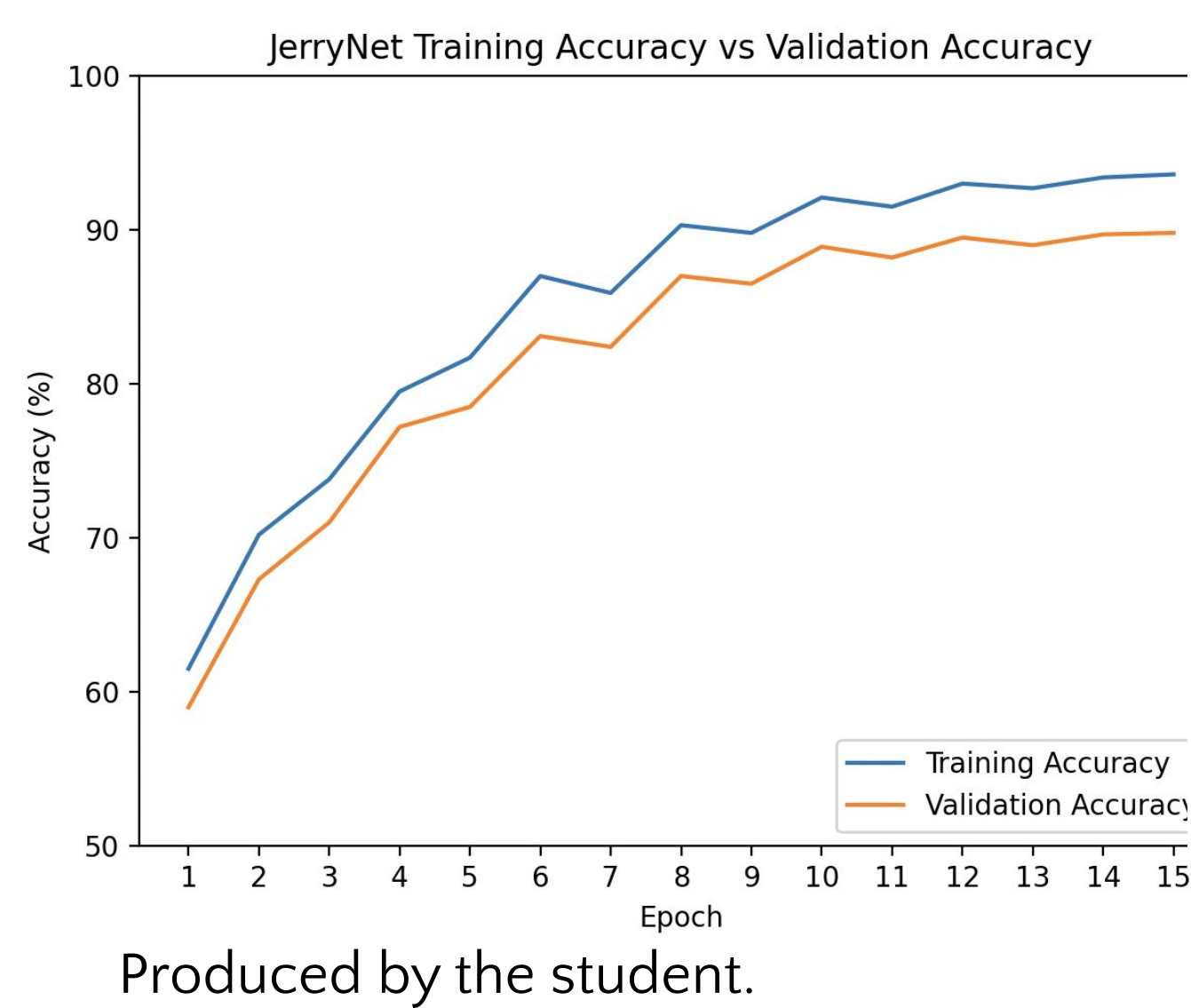
Citations

1. Sung Jin Um, Dongjin Kim and Kim, Jung Uk (2023). “Audio-Visual Spatial Integration and Recursive Attention for Robust Sound Source Localization”, <https://arxiv.org/abs/2308.06087>.
2. Yusong Wu, Ke Chen, Tianyu Zhang, Yuchen Hui, Marianna Nezhurina, Taylor Berg-Kirkpatrick, Shlomo Dubnov (2024.) “Large-scale Contrastive Language–Audio Pretraining with Feature Fusion and Keyword-to-Caption Augmentation”, <https://arxiv.org/abs/2211.06687>.

Methods: System Architecture and Models



Results



Model	Accuracy (%)	F1-Score
JerryNet	91.1	0.910
VGG16	83.3	0.832
VGG19	84.5	0.843
ResNet50V2	85.3	0.850
DenseNet121	87.0	0.867
BAST	88.2	0.880

Table II: Performance for Different CNN
Produced by the student.

JerryNet: The left image displayed the training and validation accuracy for JerryNet over 15 epochs. Using the data from the Table II, a one-way ANOVA test followed by Tukey's HSD post hoc analysis can be conducted to demonstrate a significant difference between JerryNet and other baseline models ($p < 0.001$). This indicates that JerryNet can more effectively captures the time dependency in the four audio data. leading to an accurate DoA identification in a variety of sound environments.

Audio Classification Model: The CLAP mode achieved an accuracy of 98.5% for my custom collected dataset and 95% accuracy for a 5000 dataset from AudioSet. The high performance of the fine-tuned CLAP model in zero-shot audio classification provides high reliability for real-time usage.

Metric	Our System	Baseline (LVS)
cIoU	0.892	0.876
AUC	0.658	0.641

Table III: Performance of Multimodal Integration Model

Multimodal Integration Model: As shown in Table III, our system achieved a cIoU of 0.892 and an AUC of 0.658, outperforming most of the other audiovisual sound localization models. This implies that the multimodal integration model is capable of accurately identifying the exact sound source within images because it combines the highly accurate audio visual localization model with the reliable fine-tuned YoloV9, truly unleashing the power of combining three different modes.

Discussion

These results confirm AudioAware offers a viable solution to an assistant technology capable of providing situational awareness for the deaf and hearing impaired. This bridges the gap between frontier research in artificial intelligence with assistant devices for the auditory impaired. AudioAware provides a foundation for many novel scientific innovations and societal applications in new fields—specifically for future research to develop a new generation of assistive technology.

Future Goals

- A larger and more diverse set of data should be collected for Model 1 (only 450 datasets), which can greatly enhance accuracy and improve the device.
- An optimized version of each component could be developed to reduce latency and dependence on network connectivity and computational speed on the central server, providing more reliable real-time performance.
- Reinforcement learning can be incorporated to establish an adaptive important classifier versatile to user preference over time. The current sound priority is listed in Table I.
- A more robust user interface can be established from the prototype.

Relative Priority	Class Name
1	Siren
2	Car Honking
3	Bike Bell
4	Person Talking
5	Doorbell
6	Phone Ringing
7	Dog Barking
8	Instruments

Table I: Examples of Sound Priority

Conclusion

Our study confirms that integrating deep learning models with audio, visual, and textual data significantly enhances the task of sound localization and sound classification for assistive devices. Each of our three components performs either with high accuracy or statistically greater than other baseline models.

These results confirm that our designed system indeed answers the research question, offering a viable solution to an assistive technology capable of providing situational awareness for the deaf and hearing impaired. However, recognizing the limitations of the study, there is much potential for future research to develop a new generation of assistive technologies capable of changing lives.